



Fig. 3: Placement of electrodes

Hz, found in the motor cortex, and their amplitude fluctuates based on physical movement or intent to move. Early BCI experiments, aimed at moving the cursors on a computer screen, exploited the Beta and Mu waves.

The BCI system

Overall, a BCI has four main components:

1. A signal capture system
2. A signal processing system
3. A pattern recognition system
4. A device control system

Let us look at the example of moving the mouse cursor. The signal capture system makes use of scalp electrodes (in older systems), together with a sticky gel, to obtain signal. As these electrodes capture the electrical activity (here, capturing means that some of the current passing through the axon escapes out), the signal passes through a unity gain amplifier to boost its level for processing. As these signals are impulses at discrete frequencies, we can use Fast Fourier transform algorithms to convert them to a discrete signal (basically an A/D operation). Then, in the machine or computer, DSP software is used for proper sequence generation, after which its output is fed to a pattern recognition system to select the proper pattern or symbol. In our case, it is the pointer; and then the control device produces the operation as per the signal processing and

pattern recognition results. So, we have the sequence as shown in Fig. 2.

These brain waves or signals are generated due to the differences in electrical properties carried by ions on the membrane of each neuron. The problem is that the signals are mostly blocked while only a few escape—depending on our level of concentration. More concentration means more alertness (to the task), which increases the Beta waves that can be easily detected.

If a person has hearing problem in one ear, it could be because the sensor is damaged. But since the auditory nerves are operating, a device called the cochlear implant is inserted into the ear to process sound waves, converting them to electrical signals that are then passed via the auditory nerves.

Placement of electrodes in the signal capture system

For placement of electrodes, the 10-20 system is used, which is an internationally recognised method that describes the exact location of the scalp electrodes. If we use fewer electrodes, then the distance between adjacent electrodes is 20 per cent of the total front-back or right-left distance of the skull. If we use more electrodes, then this distance is 10 per cent of the total front-back or right-left distance of the skull.

In Fig. 3, A1 and Az represent the ears' positions. There are no central electrodes; C3, Cz and C4 are marked just for geometrical indication.

But now all of this is outdated. Due to limited portability and increased interference, electrodes are no longer used for commercial applications. Currently, the method to study the electrical activity in the brain is 'Emotive EPOC', which makes use of a wireless headset. It can track conscious as well as unconscious and emotional behaviour. EPOC marks the beginning of an

Frequency Range

Wave	Hz	State of mind
Delta	Below 4	Deep sleep
Theta	4-8	Drowsiness
Alpha	8-14	Relaxed but alert
Beta	14-30	Highly alert and focused



Fig. 4: Wireless headsets

important era. It is highly precise and gives fast response.

Applications of BCI

1. The BCI can be applied not only to brain-related products but also to bionic devices. These include bionic leg, bionic eye, bionic arm, bionic spinal cord, etc. All of these are based on the same concept of neuron electrical activity to give people a new lease of life.

2. The BCI has revolutionised the gaming experience too. There are a good number of BCI-based gaming options available in the market. Normally, the 'conversation process' through the BCI is not very efficient, but recent research has resulted in new methods that are available commercially, although at higher prices.

3. The concept of virtual reality has fired the imagination of many in the BCI field, as shown in the film 'Surrogates' starring Bruce Willis.

4. There is also the concept of 'brain painting' though its efficiency, as of now, is not very high.

5. The most important applications are expected to be in relation to cloud computing. What if, through BCI, we are constantly connected to the Internet via the cloud? This can change the world in more ways than can be imagined today. ●

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